

OpenClaw Proactive Interaction

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1. Course Content

Course Overview

- Based on skills, write response strategies for specific scenarios so that OpenClaw can proactively provide solutions when facing unexpected situations
- This lesson is a solution design course for application scenarios. You need to first complete the prerequisite courses [OpenClaw Basics], [OpenClaw Interaction Methods], and [OpenClaw Skills Development]

2. Preparation

- First study the [OpenClaw Skills Development] chapter to learn how to create new skills

```
ros2 launch microros_v2_bringup openclaw_service.launch.py
```

[illegible]

3. Creating a New Skill

- Refer to [OpenClaw Skills Development] to create a response strategy skill for special situations and enable the skill

```
---
name: production-line-material-shortage-handling-process
description: Provides a preset handling process when facing material shortage
tasks during inspection tasks
version: 1.0.1
---
```

Disposal Plan for Color Material Shortage in the Production Workshop

- Proactively provide solutions to the user by asking questions; the corresponding response strategy must be executed only after user confirmation
- For example, when a color material is observed to have fewer than 2 units, it is determined that the material is short

Plan 1: Replenish from the raw material warehouse shelves

Plan 2: Go to the storage area and replenish with a different color material of the same specification

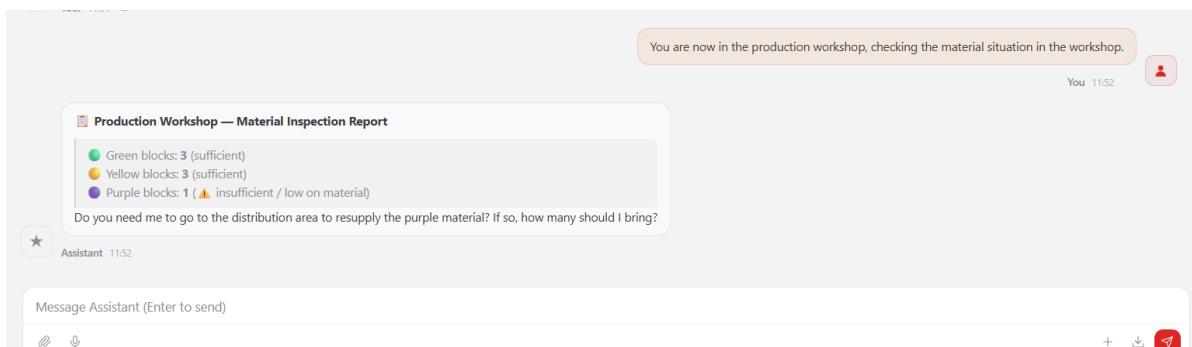
- After creation, restart the OpenClaw gateway for it to take effect

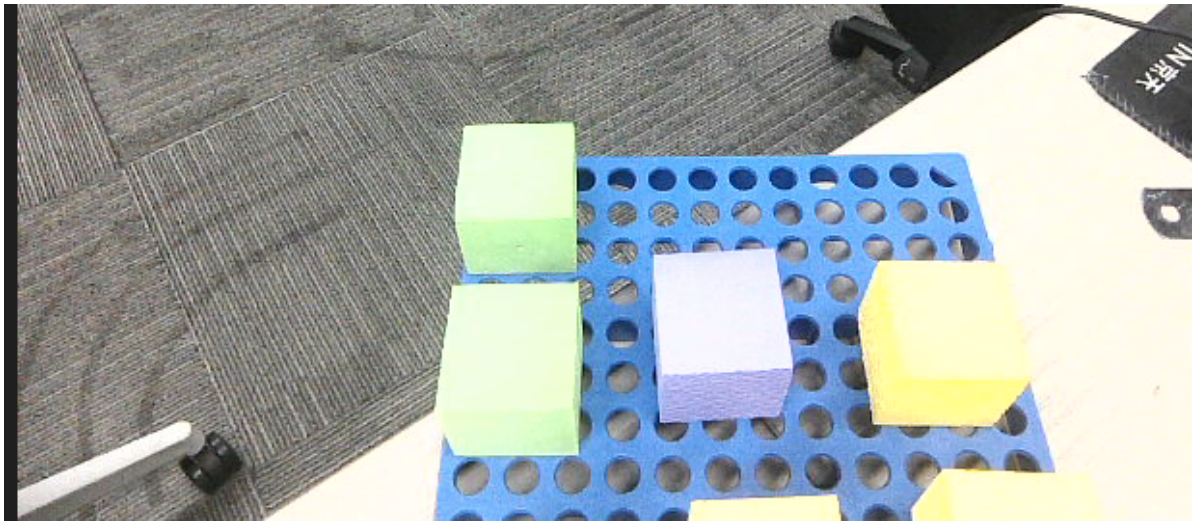
```
openclaw gateway restart
```

4. Case Demo

[!NOTE]

- You can write the command according to your own needs. The following is a case demo. Choose any interaction method; here the web WebChat is used as an example
- Here, OpenClaw proactively gives disposal suggestions based on the actual observed situation combined with the skill requirements. In the same situation, the user does not need to repeat the instruction every time





5. Extended Applications

OpenClaw's proactive interaction capability can be applied to a variety of scenarios. The following provides several reference custom solutions:

5.1 Equipment Fault Warning and Handling

```
---
name: equipment-fault-handling-process
description: Proactively provides diagnosis and handling suggestions when
  abnormal equipment operating parameters are detected
version: 1.0.0
---
```

Equipment Abnormality Proactive Warning Plan

- Monitor equipment operating status in real time and proactively alert the user when abnormalities are detected
- Provide graded handling suggestions: minor abnormality alerts, severe abnormality shutdown suggestions
- The corresponding handling strategy must be executed only after user confirmation

Handling Plan 1: Parameter Calibration

- Check whether the sensor readings are accurate
- Run the equipment self-test program
- Calibrate the deviation parameters

Handling Plan 2: Shutdown for Maintenance

- Immediately stop the equipment
- Notify maintenance personnel to arrive on site
- Record the fault log for later analysis

5.2 Production Progress Tracking and Reminders

```
---
name: production-progress-proactive-reminder
description: Periodically checks production task progress and proactively reminds
  the user at key milestones
version: 1.0.0
```

Production Progress Proactive Tracking Plan

- Periodically query the completion status of production line tasks
- Proactively remind the user at key milestones (such as material shortage, schedule delay, task completion)
- Provide suggestions for next steps

Reminder Strategy

- Material about to run out: remind to replenish 2 hours in advance
- Progress behind schedule: analyze the cause and provide catch-up suggestions
- Task completed smoothly: generate a report and notify relevant personnel

5.3 Multi-Condition Compound Trigger Strategy

name: compound-condition-intelligent-response

description: Combines multi-dimensional conditions such as time, environment, and equipment status to provide intelligent proactive responses

version: 1.0.0

Multi-Dimensional Condition Response Plan

- Comprehensively analyze multiple types of sensor data and business logic
- Proactively trigger a response when a specific combination of conditions is met
- Provide personalized handling suggestions

Trigger Condition Examples

- Time condition: during night shift (22:00-06:00)
- Environment condition: temperature above 35°C and humidity below 30
- Equipment condition: continuous operation for more than 8 hours

Response Strategy

- Proactively suggest equipment rest and maintenance
- Remind staff to pay attention to environmental safety
- Automatically adjust equipment operating parameters to energy-saving mode